

Tutorial 2 : Boolean algebra (Part 2)

Exercise 1:

Provide the logic diagrams of the following functions:

- $F1 = A.\bar{B}.C + A.B.\bar{C} + A.B.C$
- $F2 = \bar{X}.\bar{Y}.\bar{Z} + X.\bar{Y} + X.Y.Z$
- $F3 = A.B.C + A.B.\bar{C} + A.B.\bar{C}.D$
- $F4 = (A+B) . (A + \bar{B} + C)$

Exercise 2:

Simplify the following logical expressions :

- $h = \bar{A}.B + A.B$
- $m = (A + B) . (A + \bar{B})$
- $R = A + A.B$
- $Z = A . (A + B)$

Exercise 3:

1. Simplify the following functions algebraically

- $F1 = A.\bar{B}.C + A.B.\bar{C} + A.B.C$
- $F2 = \bar{X}.\bar{Y}.\bar{Z} + X.\bar{Y} + X.Y.Z$
- $F3 = A.B.C + A.B.\bar{C} + A.B.\bar{C}.D$

2. Draw the logic diagrams of the simplified functions. What do you notice compared to the logic diagrams of Exercise 1?